Functional Annotation of ExaE (PedR2) — *Pseudomonas putida* KT2440

Gene: `exaE` (synonym **`pedR2`**) · **Locus:** `PP_2672` · **UniProt:** Q88JH7 · **Length:** 212 aa

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1. Summary (Answer to the Research Question)

ExaE is not an enzyme, transporter, or structural protein — it is a DNA-binding signal-transduction protein. Specifically, it is the **response-regulator (output) component of a bacterial two-component regulatory system**, belonging to the **LuxR/NarL (FixJ) subfamily** of response regulators. In *Pseudomonas putida* KT2440 the gene annotated `exaE` (PP_2672) has been **directly characterized experimentally as `pedR2`**, the response regulator that, together with its cognate membrane sensor histidine kinase **PedS2 (PP_2671)**, forms the **PedS2/PedR2 two-component system** (Wehrmann et al., 2018, P 30158283).

Its **primary molecular function** is to work as a **phosphorylation-controlled transcription factor**. When phosphorylated on a conserved receiver-domain aspartate (Asp53) by PedS2, ExaE/PedR2 **binds promoter DNA in the cytoplasm** and exerts **dual, opposite regulatory effects**: it **activates transcription of `pedE`** (PP_2674, a Ca^{2+}-dependent periplasmic PQQ-alcohol dehydrogenase) and **represses transcription of `pedH`** (PP_2679, a lanthanide-dependent PQQ-alcohol dehydrogenase). This dual output implements the **"rare-earth-element (REE) switch"** that lets the cell choose which of two functionally redundant periplasmic alcohol dehydrogenases to deploy, depending on lanthanide availability. The pathway ultimately drives **periplasmic oxidation of a broad range of volatile alcohols and aldehydes** used as carbon/energy sources.

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2. Protein Identity and Verification

The target was rigorously verified as the intended protein:

Property	Value	Source
UniProt accession	Q88JH7	UniProtKB
Gene name	`exaE` (= `pedR2`)	UniProt / EMBL AAN68280.1; P 30158283
Locus tag	PP_2672	UniProt
Organism	*Pseudomonas putida* KT2440 (taxid 160488)	UniProt
Length	212 aa	UniProt
Architecture	N-terminal REC receiver domain + C-terminal LuxR-type HTH	UniProt / InterPro / Pfam

Identity is unambiguous and confirmed by experiment. The gene symbol `exaE` derives from orthology to the *P. aeruginosa* ethanol-oxidation regulator ExaE/EraR. However, the *P. putida* locus PP_2672 has itself been studied directly and is referred to in that primary literature as **PedR2** (Wehrmann et al., 2018, P 30158283). Both names refer to the same protein; there is no cross-organism gene-symbol confusion — the domain architecture, genomic context, and experimental data all agree.

Domain architecture (structural/bioinformatic evidence)

From UniProtKB (Q88JH7) and InterPro/Pfam: - **Residues 2–118 — Response-regulator receiver (REC) domain** (Pfam **PF00072** `Response_reg`; CheY-like superfamily IPR011006; NreC/VraR/RcsB-like REC IPR058245). Contains the **phospho-acceptor site Asp53** (annotated "4-aspartylphosphate"), the canonical residue that receives the phosphoryl group in two-component signalling. - **Residues 141–206 — LuxR-type helix-turn-helix (HTH) DNA-binding effector domain** (Pfam **PF00196** `GerE`; IPR000792 `Tscrpt_reg_LuxR_C`; IPR016032 signal-transduction response-regulator C-terminal effector). - **GO annotations:** DNA binding; transcription cis-regulatory region binding (GO:0000976); phosphorelay signal transduction system (GO:0000160); regulation of DNA-templated transcription (GO:0006355). - **No transmembrane segments** — ExaE/PedR2 is a soluble, two-domain cytoplasmic protein.

This two-domain "receiver + LuxR-HTH" layout is the textbook architecture of a **NarL/FixJ-family DNA-binding response regulator**, and the literature independently classifies ExaE/PedR2 as a member of the **LuxR family** (Mern et al., 2010, 20093290; Wehrmann et al., 2018, 30158283).

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3. Genomic Context (Evidence for Function by Synteny)

Direct UniProt neighborhood analysis (organism taxid 160488) places PP_2672 in the **`ped` alcohol-oxidation gene cluster**:

Locus	UniProt	Length	Product	Role
PP_2671	Q88JH8	424 aa	**Sensor histidine kinase (PedS2 / "ExaD")** — HAMP (161–213) + His-kinase (236–424) domains, membrane-anchored with periplasmic sensor	Cognate kinase of ExaE
PP_2672	**Q88JH7**	**212 aa**	**Response regulator ExaE / PedR2** (this study)	**DNA-binding transcription factor**
PP_2673	Q88JH6	219 aa	Pentapeptide-repeat protein	—
PP_2674	Q88JH5	631 aa	**Quinoprotein (PQQ) alcohol dehydrogenase PedE**	Regulatory target (activated)
PP_2679	—	—	**Lanthanide-dependent PQQ-ADH PedH**	Regulatory target (repressed)

The immediate adjacency of the **cognate kinase (PP_2671)**, the **regulator (PP_2672)**, and the **structural enzyme gene `pedE` (PP_2674)** mirrors the syntenic `exaD–exaE–exaA` arrangement in *P. aeruginosa*, providing strong structural/evolutionary support for the regulator→dehydrogenase relationship.

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4. Primary Function — Mechanistic Detail

4.1 ExaE/PedR2 is the output of the PedS2/PedR2 two-component system

Two-component systems couple a **sensor histidine kinase** to a **response regulator** via His→Asp phosphotransfer. Here: - **PedS2 (PP_2671)** is a **membrane-bound sensor kinase** (HAMP + kinase domains) that senses **lanthanide (rare-earth-element, La^{3+}) availability**, most likely by direct La^{3+} binding at its **periplasmic region** (Wehrmann et al., 2018, P 30158283).
- **PedR2 (PP_2672 = ExaE)** is the **cognate LuxR-type response regulator**. Phosphotransfer to its **Asp53** switches it into the active, DNA-binding state.

4.2 Dual transcriptional output — the "REE switch"

The defining experimental result (adaptive evolution, site-directed mutants, transcriptional `lacZ`/reporter fusions, complementation; Wehrmann et al., 2018, P 30158283):

- **In the absence of La^{3+}:** PedS2 is kinase-active → **phospho-PedR2 accumulates → activates `pedE`** (Ca^{2+}-dependent PQQ-ADH) **and represses `pedH`** (Ln^{3+}-dependent PQQ-ADH). The cell uses **PedE**.
- **In the presence of La^{3+}:** La^{3+} binding lowers PedS2 kinase activity → **less phospho-PedR2** → `pedE` activation falls and `pedH` repression is relieved; combined with activation of `pedH` by a second, still-undefined regulatory module, the cell **switches to PedH**.

Thus **ExaE/PedR2 is a dual-function transcription factor — a positive regulator of `pedE` and a negative regulator of `pedH`** — and its phosphorylation status is the molecular variable that selects between the two enzymes. This "inverse regulation" of two functionally redundant enzymes is an adaptive strategy to optimize growth on volatile alcohols according to environmental lanthanide supply (Wehrmann et al., 2017, P 28655819).

4.3 What the pathway accomplishes (the enzymes ExaE controls)

The downstream enzymes are **periplasmic PQQ-dependent alcohol dehydrogenases (PQQ-ADHs / PQQ-EDHs)**: - **PedE (PP_2674):** Ca^{2+}-dependent PQQ-ADH. - **PedH (PP_2679):** Ln^{3+}-dependent PQQ-ADH. Both oxidize a **broad and overlapping substrate range — linear and aromatic primary and secondary alcohols, as well as aldehydes** — and are **crucial for efficient growth on volatile alcohols** (Wehrmann et al., 2017, P 28655819).

These type-II quinoprotein ADHs are **soluble periplasmic enzymes** that pass electrons to a periplasmic cytochrome *c* (Toyama et al., 2004, P 15234265).

ExaE/PedR2's biological "job" is therefore to **switch on the correct periplasmic alcohol-oxidation machinery** for carbon/energy acquisition and detoxification of volatile organic compounds.

4.4 Evolutionary/ortholog context

ExaE/PedR2 is the ortholog of the *P. aeruginosa* response regulator **ExaE (renamed EraR)** of the **EraSR (formerly ExaDE)** two-component system, which activates transcription of the quinoprotein ethanol dehydrogenase gene `exaA` (QEDH) during aerobic ethanol oxidation (Schobert & Görisch, 2001, P 11158353; Görisch, 2003, P 12686116; Mern et al., 2010, P 20093290).

In *P. aeruginosa* the cognate kinase ExaD/EraS is **soluble/cytoplasmic**, whereas in *P. putida* PedS2 is **membrane-bound with a periplasmic La^{3+} sensor** — a mechanistic divergence, though the regulator's core role (LuxR-type activator of a PQQ-alcohol dehydrogenase gene) is conserved.

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5. Localization — Where ExaE Acts

- **ExaE/PedR2 acts in the cytoplasm** as a DNA-binding transcription factor (soluble two-domain protein, no transmembrane segments; C-terminal LuxR-HTH binds promoter DNA).
- **Signal reception occurs at the membrane/periplasm:** the cognate kinase PedS2 senses lanthanides at its periplasmic region and relays the signal across the membrane via His→Asp phosphotransfer.
- **The catabolic output occurs in the periplasm:** the PedE/PedH PQQ-ADHs are soluble periplasmic enzymes (Toyama et al., 2004, P 15234265; Wehrmann et al., 2017/2018).

There is thus a clean spatial division of labour: **periplasmic sensing (PedS2) → cytoplasmic transcriptional decision (ExaE/PedR2) → periplasmic enzymatic oxidation (PedE/PedH).**

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6. Supported and Refuted Hypotheses

Supported: - ExaE is a **LuxR/NarL-family DNA-binding response regulator** (domain architecture + literature). - ExaE = **PedR2**, the response regulator of the **PedS2/PedR2 TCS** (direct experiment; P 30158283).
- ExaE is a **dual regulator**: **activates** `pedE`, **represses** `pedH` (P 30158283).
- The system mediates the **lanthanide-dependent "REE switch"** between two periplasmic PQQ-ADHs (P 30158283; P 28655819).

- ✅ ExaE acts as a **cytoplasmic transcription factor**; downstream enzymes are **periplasmic** (P 15234265 P 28655819).

Refuted / corrected: - ❌ Initial (orthology-based) assumption that the cognate kinase is soluble/cytoplasmic. **Corrected:** in *P. putida* the cognate kinase **PedS2 is membrane-bound with a periplasmic La^{3+}-sensing region** (P 30158283). (The soluble-kinase description is true for the *P. aeruginosa* ortholog ExaD, not for *P. putida* PedS2.) - ❌ That ExaE is an enzyme/transporter (its function is purely regulatory/DNA-binding).

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7. Limitations and Future Directions

- **Direct DNA-binding evidence:** the exact `pedE` / `pedH` operator sequences bound by phospho-PedR2 have not been mapped by footprinting/EMSA in the primary abstracts reviewed; the activation/repression roles are inferred from genetics and reporter fusions. In-vitro binding-site definition would strengthen the mechanism.
- **`pedH` activation module:** a second, "yet unknown" regulator that positively controls `pedH` in the presence of La^{3+} remains unidentified (P 30158283).
- **Phospho-Asp53 requirement:** Asp53 is the predicted/annotated phospho-acceptor; a D53A/D53E mutant phenotype would formally confirm phosphorylation-dependence in *P. putida*.
- **Additional regulon members:** whether phospho-PedR2 controls genes beyond `pedE` / `pedH` (e.g., PQQ biosynthesis, cytochrome, downstream aldehyde dehydrogenases) is not fully defined.

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8. Key References

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mSphere 3:e00376-18 (2018). PMID **30158283**. *(Direct characterization of PP_2671/PP_2672 = PedS2/PedR2; dual activation of pedE and repression of pedH.)*

2. Wehrmann M, Billard P, Martin-Meriadec A, Zegeye A, Klebensberger J. **Functional Role of Lanthanides in Enzymatic Activity and Transcriptional Regulation of PQQ-Dependent Alcohol Dehydrogenases in *P. putida* KT2440.** *mBio* 8:e00570-17 (2017). PMID **28655819**. *(PedE = PP_2674 Ca^{2+}-dependent, PedH = PP_2679 Ln^{3+}-dependent; broad alcohol/aldehyde substrate range; periplasmic; inverse regulation.)*
3. Wehrmann M, Toussaint M, Pfannstiel J, Billard P, Klebensberger J. **The Cellular Response to Lanthanum Is Substrate Specific…** *mBio* 11:e00516-20 (2020). PMID **32345644**.
4. Mern DS, Ha SW, Khodaverdi V, Gliese N, Görisch H. **A complex regulatory network controls aerobic ethanol oxidation in *P. aeruginosa*…** *Microbiology* 156:1505–1516 (2010). PMID **20093290**. *(ExaE renamed EraR; LuxR family; EraSR = ExaDE cognate pair.)*
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6. Schobert M, Görisch H. **A soluble two-component regulatory system controls expression of QEDH…** *Microbiology* 147:363–372 (2001). PMID **11158353**. *(Identification of the ExaD/ExaE TCS controlling exaA promoter.)*
7. Toyama H, Mathews FS, Adachi O, Matsushita K. **Quinohemoprotein alcohol dehydrogenases: structure, function, and physiology.** *Arch Biochem Biophys* 428:10–21 (2004). PMID **15234265**. *(Type-II PQQ-ADHs are soluble periplasmic enzymes.)*
